RESUME SCREENING USING NLP

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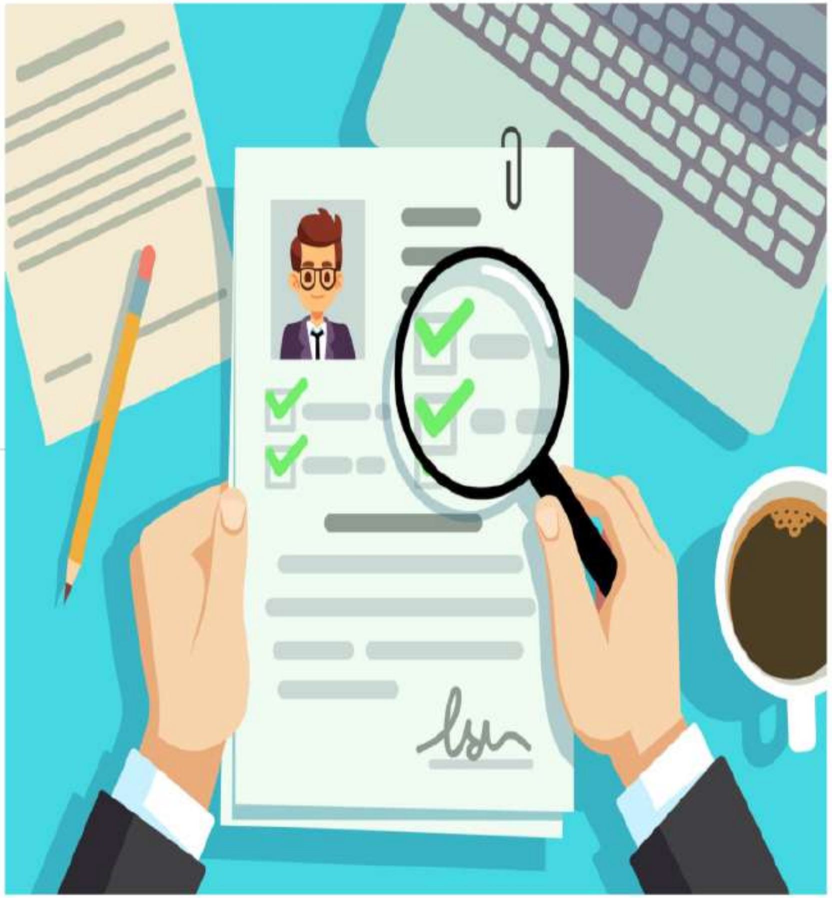
ABSRACT

In the modern job market, organizations receive an overwhelming number of resumes for every job opening.Recruiters are expected to analyze them within a limited time, which makes manual screening impractical.As a result, many qualified candidates are often overlooked, while the hiring process becomes slow and biased.This situation clearly highlights the need for an intelligent system that can assist recruiters in decision-making.



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Problem Statements

❗ **High Volume & Limited Time:** Recruiters spend an average of just 7 seconds per resume, often missing qualified candidates amidst the noise of hundreds of applications.

📄 **Data Complexity**: Resumes arrive in varied formats (PDF, DOCX) with non-standard layouts, making data extraction difficult for traditional keyword parsers.

⚖ **Bias & Inconsistency:** Manual screening is susceptible to unconscious bias, while rigid keyword matching creates false negatives, compromising the quality and fairness of hiring.

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NLP ENGINE: HOW IT WORKS

Natural Language Processing (NLP) transforms unstructured resume data into structured insights. By simulating human reading comprehension, the AI engine processes documents through five distinct stages to ensure accurate candidate matching.

INGEST & PARSE

Extracts raw text from diverse formats (PDF, DOCX) and identifies document structure.

CLEAN & NORMALIZE

Removes noise, corrects formatting errors, and standardizes terminology.

JOB MATCHING

scores against job descriptions to rank candidates.

EXTRACT ENTITIES

Identifies key entities: Skills, Job Titles, Education, and Years of Experience.

SEMANTIC MEANING

Uses vector embeddings to understand context beyond simple keywords.

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BENEFITS OF NLP-BASED RESUME SCREENING

OPERATIONAL EXCELLENCE

- **Speed & Scale:** Automates the first-pass screening of thousands of resumes instantly, reducing time-to-shortlist by up to 75%.
- **Consistency:** Applies standardized criteria to every candidate, eliminating human fatigue and reducing variability in assessment.
- **Accuracy**: Semantic matching identifies qualified candidates even when keywords don't match exactly, improving candidate recall.

FAIRNESS & EXPERIENCE

- **Bias Reduction:** Focuses on skills and experience rather than demographics, mitigating unconscious bias in the early funnel.
- **Candidate Experience**: Enables faster feedback loops and provides clearer role fit indications, reducing candidate drop-off.
- **Compliance & Insight:** Creates traceable decision logs for auditability and provides deep analytics on talent pipeline quality.

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KEY NLP TECHNIQUES

Named Entity Recognition (NER)

Identifies and categorizes key entities such as candidate names, educational institutions, organizations, and specific dates.

Keyword & Phrase Extraction

Utilizes TF-IDF and KeyBERT to isolate core skills and competencies from unstructured text.

Semantic Embeddings

Leverages models like BERT to understand context and synonyms (e.g., matching "ML" with "Machine Learning").

Classification & Normalization

Categorizes resumes by role suitability and maps varied job titles to a standardized ontology.

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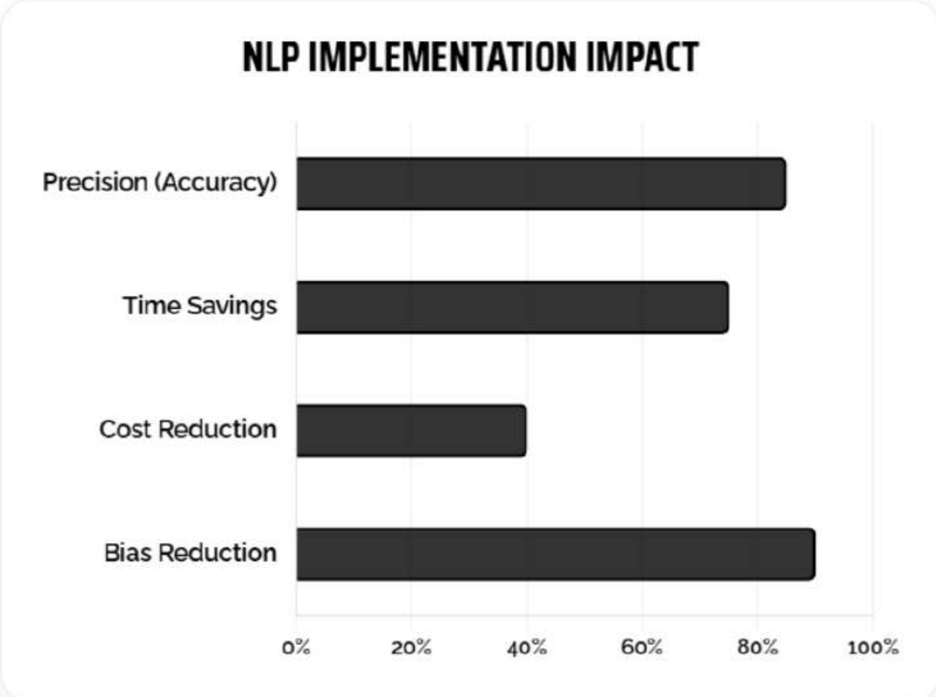
APPLICATIONS & SUCCESS METRICS

NLP transforms recruitment by seamlessly integrating with ATS platforms to automate shortlisting, enabling precise semantic talent matching against job descriptions, and proactively rediscovering candidates from past pools.

- ATS Integration & Auto-Ranking
- Talent Rediscovery & Contextual Search
- Fairness KPIs & Bias Monitoring
- Drastic Time-to-Hire Reduction



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LITERATURE SURVEY

Earlier research focused on keyword-based resume screening, which produced inaccurate results.
Later studies introduced machine learning models that improved classification but required large training datasets.
Recent research emphasizes NLP-based approaches, as they provide better understanding of resume content and semantic meaning.
These studies prove that NLP-based systems are more effective than traditional screening methods.

-Early systems used **keyword-based resume filtering**
-Keyword methods lacked **context understanding**
-Machine learning improved **resume classification**
-ML models required **large training data**
-NLP-based approaches analyze **resume text effectively**
-Techniques like **TF-IDF and similarity matching** are used
-NLP systems provide **higher accuracy and reduced bias**

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Identified Gaps in Current Approaches

Despite advancements, existing systems still suffer from several limitations.
Most systems lack contextual understanding and depend heavily on predefined keywords.
They struggle to handle diverse resume formats and fail to adapt to new or emerging skills.
These gaps indicate the need for a more flexible and intelligent resume screening system

CONCLUSION & FUTURE SCOPE

This project demonstrates how NLP can revolutionize the recruitment process by automating resume screening.
It improves efficiency, fairness, and accuracy in candidate selection.
In the future, this system can be enhanced using deep learning models, multilingual resume analysis, and AI-based interview recommendations.